[redacted] Chat Log

Transformation Code

```
SELECT
  GENDER,
  AGE,
  'SMOKING' AS variable,
  SMOKING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'YELLOW_FINGERS' AS variable,
  YELLOW_FINGERS AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'ANXIETY' AS variable,
  ANXIETY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'PEER_PRESSURE' AS variable,
  PEER_PRESSURE AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'CHRONIC_DISEASE' AS variable,
  CHRONIC_DISEASE AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'FATIGUE' AS variable,
  FATIGUE AS value
```

```
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'ALLERGY' AS variable,
  ALLERGY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'WHEEZING' AS variable,
  WHEEZING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'ALCOHOL_CONSUMING' AS variable,
  ALCOHOL_CONSUMING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'COUGHING' AS variable,
  COUGHING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'SHORTNESS_OF_BREATH' AS variable,
  SHORTNESS_OF_BREATH AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'SWALLOWING_DIFFICULTY' AS variable,
  SWALLOWING_DIFFICULTY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
```

```
  'CHEST_PAIN' AS variable,
  CHEST_PAIN AS value
FROM dataset
```

Derivation Dialog

user

[CONTEXT]

[TABLE dataset]

Column metadata:

GENDER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ["('F',)", "('M',)"]}
AGE (BIGINT) ---- {'count': 3000, 'unique_count': 51, 'null_count': 0, 'sample_values': ['(41,)', '(46,)', '(56,)', '(30,)', '(60,)', '(74,)', '(35,)']}
SMOKING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
YELLOW_FINGERS (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ANXIETY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
PEER_PRESSURE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
CHRONIC_DISEASE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
FATIGUE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ALLERGY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
WHEEZING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ALCOHOL_CONSUMING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
COUGHING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
SHORTNESS_OF_BREATH (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
SWALLOWING_DIFFICULTY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
CHEST_PAIN (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
LUNG_CANCER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ["('NO',)", "('YES',)"]}

Sample data:

| GENDER | AGE | SMOKING | YELLOW_FINGERS | ANXIETY | PEER_PRESSURE | CHRONIC_DISEASE |
FATIGUE | ALLERGY | WHEEZING | ALCOHOL_CONSUMING | COUGHING | SHORTNESS_OF_BREATH |
SWALLOWING_DIFFICULTY | CHEST_PAIN | LUNG_CANCER |
0| M | 65 | 1 | 1 | 1 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | NO |
1| F | 55 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | NO |
2| F | 78 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | YES |
3| M | 60 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 2 | YES |
4| F | 80 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 1 | 1 | 2 | NO |

[GOAL]

Make a complete EDA of my dataset

[OUTPUT]

assistant

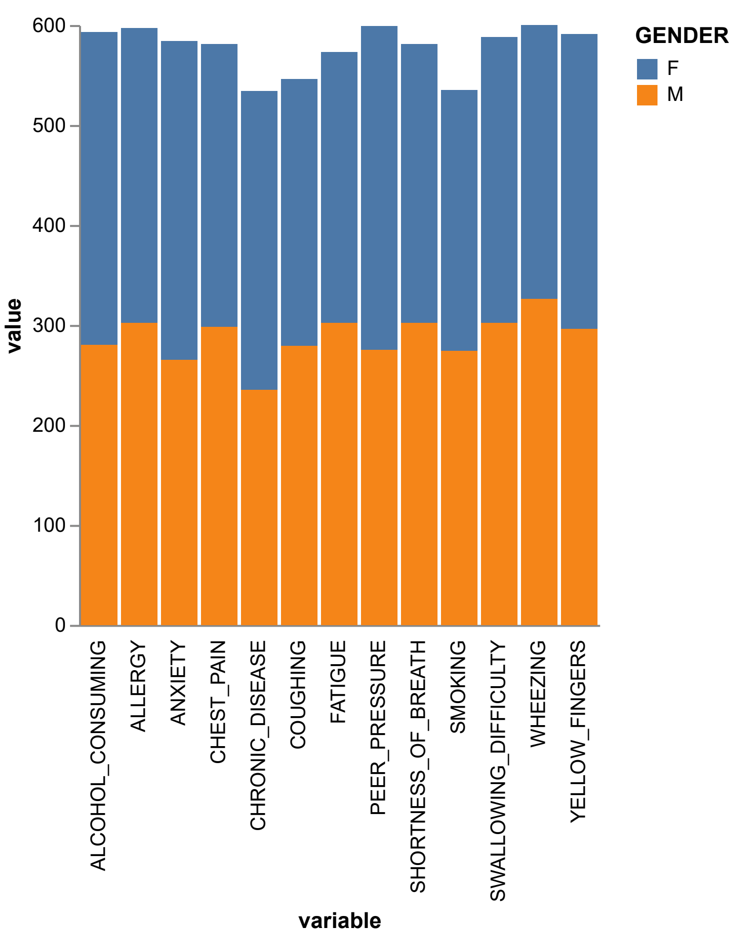
```json
{
  "mode": "overview",
  "recommendation": "The dataset appears to be related to lung cancer and various health factors. To perform a complete EDA, we will first reshape the data from wide format to long format to make it easier to analyze and visualize.",
  "output_fields": ["GENDER", "AGE", "variable", "value"],
  "chart_type": "bar",
  "visualization_fields": ["variable", "value", "GENDER"]
}
```

```sql
SELECT
  GENDER,
```

```
  AGE,
  'SMOKING' AS variable,
  SMOKING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'YELLOW_FINGERS' AS variable,
  YELLOW_FINGERS AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'ANXIETY' AS variable,
  ANXIETY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'PEER_PRESSURE' AS variable,
  PEER_PRESSURE AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'CHRONIC_DISEASE' AS variable,
  CHRONIC_DISEASE AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'FATIGUE' AS variable,
  FATIGUE AS value
FROM dataset
```

```
UNION ALL
SELECT
  GENDER,
  AGE,
  'ALLERGY' AS variable,
  ALLERGY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'WHEEZING' AS variable,
  WHEEZING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'ALCOHOL_CONSUMING' AS variable,
  ALCOHOL_CONSUMING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'COUGHING' AS variable,
  COUGHING AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'SHORTNESS_OF_BREATH' AS variable,
  SHORTNESS_OF_BREATH AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
```

```
  'SWALLOWING_DIFFICULTY' AS variable,
  SWALLOWING_DIFFICULTY AS value
FROM dataset
UNION ALL
SELECT
  GENDER,
  AGE,
  'CHEST_PAIN' AS variable,
  CHEST_PAIN AS value
FROM dataset
```



AI generated results can be inaccurate, inspect it!

sample data from **view_oxcr** (4 columns × 39000 rows) show column stats

GENDER	AGE	variable	value
M	65	SMOKING	1
F	55	SMOKING	1
F	78	SMOKING	2
M	60	SMOKING	2
F	80	SMOKING	1
F	58	SMOKING	1
F	70	SMOKING	1
F	74	SMOKING	2
M	77	SMOKING	1
......	......	......	......